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Bio 301

Past Papers Solved 2020

Day 1st

1. **Monosomy?**

Monosomy is the state of having a single copy of a chromosome pair instead of the usual two copies found in diploid cells

2. **Causes of monosomy:**

Followings are the causes of monosomy

Non disjunction – one gamete receives two copies of homologous chromosomes and other will have no copy.

Loss of chromosomes.

As it moves towards pole of cell during anaphase.

Fruit fly monosomy, is the cause of some human diseases such as Turner syndrome and Cri du Chat syndrome

3. **What are structural abnormalities?**

Structural abnormalities are when part of an individual chromosome is missing, extra, switched to another chromosome, or turned upside down.

4. **Types based on genetic material**

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Balanced – chromosome complement is complete.

Unbalanced – when there is incorrect amount of genetic material.

Types

- Translocations
- Deletions
- Insertions
- Inversions
- Ring chromosomes
- Iso chromosome

5. **point deletions**

For point deletions, one nucleotide has been deleted from the sequence.

6. **Suppressor mutation**

Suppresses the effect of other mutation Occurs at a site different from the site of original mutation.

7. **Intragenic suppression**

Intragenic suppression results from suppressor mutations that occur in the same gene as the original mutation.

8. **Intergenic suppression**

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Intergenic (extragenic) suppression relieves the effects of a mutation in one gene by a mutation somewhere else within the genome. Second mutation is not on same gene as original mutation

9. Penetrance and Expressivity

Penetrance measures the proportion of individuals in a population who carry a specific gene and express the related trait. Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.

In genetics, **expressivity** is the degree to which a phenotype is expressed by individuals having a particular genotype. (Alternately, it may refer to the expression of particular gene by individuals having a certain phenotype.)

10. Low and full Penetrant

Low Penetrance An allele with low penetrance will only sometimes produce the symptom or trait with which it has been associated at a detectable level. In low penetrance, it is difficult to distinguish environmental from genetic factors.

Highly Penetrant If an allele is highly penetrant, and then the trait it produces will almost always be apparent in an individual carrying the allele.

11. Polyploidy

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Polyploidy is the heritable condition of possessing more than two complete sets of chromosomes

Two major types

- Auto polyploidy
- Allopolyploid

12. Frequency of autosomal recessive disorders

Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

13. Trisomy

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two.

Trisomy is a type of aneuploidy. Trisomy of chromosome 21, which is found in Down syndrome, is called trisomy 21 inversion mutation.

14. X-linked recessive disorders

- Duchene Muscular
- Dystrophy Hunter's Disease
- Menkes Disease

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- Hemophilia A and B
- Color Blindness

15. **Diagnostic tests for genetic disorders**

Physical analysis: Physical analysis for affected individual

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks. gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus.,

16. **Polymorphism**

Polymorphism is a discontinuous genetic variation resulting in the occurrence of several different forms or types of individuals among the members of a single species

17. **Aneuploidy:** Change in chromosomes number can occur by the addition/deletion of chromosome or part of a chromosome.
18. **Euploidy:** Gain of one or more complete sets of chromosomes

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19. pattern of inheritance

Most genes have one or more versions due to mutations or polymorphisms referred to as alleles.

There are five basic modes of inheritance for single-gene diseases:

- autosomal recessive
- autosomal dominant
- X-linked dominant
- X-linked recessive
- mitochondrial.

20. Multifactorial diseases six:

Genetic disorders controlled by many genes plus the effects of the environment

- Asthma
- Cancers
- Diabetes
- Heart diseases
- Hypertension
- Obesity

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Day 2nd

21. Pleiotropy

Pleiotropy

The ability of a gene to affect an organism in many ways is called pleiotropy.

Pleiotropy refers to an allele which has more than one effect on the phenotype.

22. Types of deletion include the following:

- Terminal deletion – a deletion that occurs towards the end of a chromosome.
- Intercalary/interstitial deletion – a deletion that occurs from the interior of a chromosome.
- Microdeletion – a relatively small amount of deletion (up to 5Mb that could include a dozen genes)

23. Method of mapping gene disorders

• Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

24. How genetic mapping performed

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Scientists isolate DNA from samples and examine it for the unique patterns of bases. Molecular techniques are referred to as genotyping polymorphisms or markers.

25. Applications of molecular markers – short range

- Parentage determination
- Genetic distance estimation
- Determination of twin zygosity

26. Applications of Molecular Markers – Long

- Gene mapping & mapping of QTL by linkage
- Markers assisted selection of breeds

27. Function of Pedigree?

A pedigree may be used to establish the probability of a child having a particular disorder or condition.

It may be used to discover where the genes in question are located (x, y, or autosome chromosome), and to determine whether a trait is dominant or recessive.

28. Define Epigenetics?

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence

29. RAPD And ALFP Difference?

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RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

AFLP is a PCR-based tool used in genetics research, DNA fingerprinting, and in the practice of genetic engineering. Developed in the early 1990s by Key gene, AFLP uses restriction enzymes to digest genomic DNA, followed by ligation of adaptors to the sticky ends of the restriction fragments.

30. Point Mutation?

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

31. Complications Of Inheritance?

Complications to inheritance patterns:

- Multiple genes – epistasis
- polygenic traits Genes & the environment - sex-influenced traits
- incompletepenetrance.

Complications: A common recessive inheritance can give dominant pattern, Autosomal dominant inheritance with non-penetrance, Autosomal dominant inheritance with variable expression.

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32. Mutation Effects in Functioning of Genes?

Some mutations don't have any noticeable effect on the phenotype of an organism. This can happen in many situations: perhaps the mutation occurs in a stretch of DNA with no function, or perhaps the mutation occurs in a protein-coding region, but ends up not affecting the amino acid sequence of the protein.

33. Causes Of Epigenetics?

Stress Hormone Causes Epigenetic Changes. Researchers found that chronic exposure to a stress hormone causes modifications to DNA in the brains of mice, prompting changes in gene expression

34. Genetic Mapping in Humans?

Genetic mapping

Genetic mapping also called linkage mapping provides clues about which chromosome contains the gene and precisely where it lies on that chromosome.

Single gene disorders identified

 Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

How genetic mapping performed

 To produce a genetic map, researchers collect blood or tissue samples from family members where a certain disease or trait is prevalent.

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How genetic mapping performed

 Scientists isolate DNA from samples and examine it for the unique patterns of bases. Molecular techniques are referred to as genotyping polymorphisms or markers.

DNA markers help in location of genes

 Before researchers identify the gene responsible for the disease or trait, DNA markers can tell them roughly where the gene is on chromosome.

More markers increase chance of linkage

 The more DNA markers there are on a genetic map, the more likely it is that one will be closely linked to a disease gene.

Types of Genetic markers

 Several different types of genetic markers, the type most used on genetic maps is known as a microsatellite and SNPs.

Genetic mapping provides clues about location of genes

 Genetic mapping provides clues about the location of genes on chromosomes.

35. <u>Forward, Reverse, And Neutral Mutations: 3marks</u>

- **Forward mutation:** Mutation that alters the wild type phenotype into mutant phenotype.
- **Reverse mutation:** Mutation that changes a mutant phenotype back into the wild type.

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- **Neutral mutation:** Mutation that alters the amino acid sequence of the protein but does not change its function as replaced amino acid is chemically similar or the affected AA has little influence on protein function

36. Point Mutations and Enlist 3 Structural Changes in Chromosomes?

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA.

Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

- Deletion
- duplication
- translocation
- transcription

Structural Change in the Structure of Chromosomes. ... Chromosomes are the vehicle of hereditary material or genes. Any alteration, addition or deletion of chromosomal part leads to alteration of number, position or sequence of genes in the chromosome.

37. Complications in Inheritance?

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- Epistasis
- polygenic traits Genes & the environment
- sex-influenced traits
- incomplete penetrance.

38. Modification ofEpigenetic?

Epigenetic modification - non coding RNA is a new mechanism for gene regulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs).

Three common types of modification

- Cytosine methylation
- Histone's modifications
- siRNAs

39. Mutation Alter Genes Expression?

Non-coding mutations: Promoter/enhancer element trinucleotide repeats in 5 and 3'UTRs Splice sites.

Effect of mutation on gene function.

Null allele (loss of function) - no gene product.

Hypomorph – decreased activity.

Gain of function - increased activity.

Dominant negative-antagonizes normal product.

Neomorph - novel activity.

Predicting that a gene product won't do the job.

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Deletion, nonsense, frameshift of sequence is deleterious.

Mutation in splice site usually has sever effects.

Missense mutations depend on location in protein.

40. Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

Day 3rd

41. Sickle cell anemia?

Sickle cell anemia (sickle cell disease) is a disorder of the blood caused by an inherited abnormal hemoglobin (the oxygen-carrying protein within the red blood cells).

The abnormal hemoglobin causes distorted (sickled) red blood cells.

The sickled red blood cells are fragile and prone to rupture.

42. Point mutation and cause?

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

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Cause:

The protein may lose its function, which can result ina disease in the organism.

For example, sickle-cell disease is caused by a single point mutation (a missense mutation) in the beta-hemoglobin gene that converts a GAG codon into GUG, which encodes the amino acid valine rather than glutamic acid.

43. Mutation nomenclature?

Standard nomenclature based on genomic DNA reference sequences and RNA reference sequences is not shown. “Genomic DNA reference sequence” simply indicates any human DNA sequence in the database that is not based on a cDNA sequence. ... Example of standard mutation nomenclature based on a coding DNA sequence.

44. VNTR and application?

VNTR are short nucleotide sequences organized as a tandem repeat on genomes. Found on many chromosomes, Show variations in length between individuals.

Applications of VNTR

Microbiology DNA fingerprinting Genetic diversity Forensics Mapping of genomes Breeding patterns of wild or domesticated animals.

45. Single and Multifactorial Mutation?

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Single gene disorders are caused by DNA changes in one particular gene, and often have predictable inheritance patterns. Over 10,000 human disorders are caused by a change, known as a mutation in a single gene These are known as single gene disorders

Multifactorial Disorders.

Researchers are learning that nearly all conditions and diseases have a genetic component. Some disorders, such as sickle cell disease and cystic fibrosis, are caused by mutations in a single gene.

Conditions caused by many contributing factors are called complex or multi factorial disorders.

46. Autosomal Dominant Disordered?

A single abnormal gene on one of the first 22 nonsex (autosomal) chromosomes from either parent can cause an autosomal disorder. Dominant inheritance means an abnormal gene from one parent can cause disease.

Examples of autosomal dominant disorders include

- Marfan syndrome
- neurofibromatosis type 1.

47. Function Of STR?

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Short Tandem Repeat (STR) analysis is one of the most useful methods in molecular biology which is used to compare specific loci on DNA from two or more samples.

A short tandem repeat is a microsatellite, consisting of a unit of two to thirteen nucleotides repeated hundreds of times in a row on the DNA strand.

48. During DNA Extraction How Proteins Are Denatured?

Adding proteinase

49. Define Locus?

A locus (plural loci) in genetics is a fixed position on a chromosome, like the position of a gene or a marker (genetic marker). Each chromosome carries many genes; human's estimated 'haploid' protein coding genes are 19,000–20,000, on the 23 different chromosomes.

50. Sex Linked Traits and Disorder?

Genes on the X or Y chromosome are called sex-linked.

For example, an X-linked recessive allele in humans causes hemophilia, which is much more common in males than females because males are hemizygous and therefore express thetrait when they inherit one mutant allele.

51. Define Multiple Alleles with Example?

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Multiple alleles: Three or more alternative forms of a gene (alleles) that can occupy the same locus. However, only two of the alleles can be present in a single organism. For example, the ABO system of blood groups is controlled by three alleles, only two of which are present in an individual.

52. Genetic Disorder?

A genetic disorder is a genetic problem caused by one or more abnormalities in the genome. Most genetic disordersare quite rare and affect one person in every several thousands or millions. Genetic disorders may be hereditary, meaning that they are passed down from the parents' genes.

53. Genetic Mapping?

Gene mapping describes the methods used to identify the locus of a gene and the distances between genes. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome.

54. What Is Location of Intronic Sequence with Respect to Exons?

Sequences that are joined together in the final mature RNA after RNA splicing areexons. Introns are found in the genes of most organisms and many viruses, and can be located in a wide range of genes, including those that generate proteins, ribosomal RNA (rRNA), and transfer RNA (tRNA).

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55. . Is There Any Rules for Mutation Nomenclature?

Mutation Nomenclature

 While describing sequence variants, it should be written according to **HGNC rules.**

56. . Write Down At Least Three Mitochondrial Diseases?

MELAS Mitochondrial encephalomyopathy

lactic acidosis and stroke-like episodes is MELAS.

Kearns-Sayre Syndrome NARP

NARP stands for neurogenic weakness, ataxia and retinitis pigmentosa.

57. Structural change in chromosome?

Structural Change in the Structure of Chromosomes. Chromosomes are the vehicle of hereditary material or genes. Any alteration, addition or deletion of chromosomal part leads to alteration of number, position or sequence of genes in the chromosome.

58. Gene frequency in autosomal dominant traits?

Gene Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals.

59. Chromosomal non disjunction types and causes?

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Non-disjunction: When two homologous chromosomes or sister chromatids fail to separate during cell division, such a phenomenon is called as non-disjunction

Types of Non-disjunction

- Meiotic non-disjunction
- Mitotic non-disjunction

Meiotic Non-disjunction

If non-disjunction occurs during 1st meiotic division, then all the sperms/ova derived from primary spermatocyte/oocytes will be abnormal.

Failure of chromosomes separation

Failure of a pair of homologous chromosomes to separate in meiosis I. ′Failure of sister chromatids to separate during meiosis II.

Failure of sister chromatids to separate during mitosis.

Mitotic Non-disjunction

Mitotic non-disjunction results in cells with different number of chromosomes. Breaking of the spindle fibers during metaphase or anaphase. If spindle fibers do not form for one of the chromosomes, then during anaphase it will not separate into two chromatids and one cell will have an extra chromatid and the other will be lacking one. Mitotic

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nondisjunction causes trisomic and monosomic daughter cells to be formed.

Causes of non-disjunction

- Aging effect on primary oocyte
- Radiation \
- Delayed fertilization after ovulation

60. Frameshift mutation?

Frameshift mutations result in all amino acids downstream from the mutation site being completely different from wild type. These proteins are generally nonfunctional.

Day 4th

61. What is chromosomal aberration? How it is classified?

Chromosomal Aberrations:

any change in normal number or structure of chromosomes. Affects individuals may be due altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes

Classification

- Numerical (usually due to de novo error in meiosis).

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- Structural (may be due to de novo error in meiosis or inherited).
- Different cell lines (occurs post-zygotically).

Numerical changes

- Aneuploidy

  Hypo ploidy Hyper ploidy
- Euploidy

  MonoploidyDiploidy Polyploidy

Structural

- Translocations
- Deletions
- Duplications
- Inversions

Different cell lines

- Mosaicism

62. Intronic position with respect to exon?

Sequences that are joined together in the final mature RNA after RNA splicing are exons. Introns are found in the genes of most organisms and many viruses and gene can be located in a wide range of genes and included those that generate proteins and ribosomal RNA rRNA and transfer TRNA.

63. Single nucleotides polymorphism?

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Single Nucleotide Polymorphism SNP:

Genetic variations in a DNA sequence when a single nucleotide in a genome is altered

humans SNP distribution

- 25,000 non-synonymous
- 50,000 Synonymous
- 25,000 Regulatory regions
- 50,000 Intragenic
- 50,000 Distributed intergenic

SNPs Most common changes

Transitions: Purines to Purines

Pyrimines to Pyrimidines

Transversion: Purines to Pyrimidines

Pyrimidines to Purines

SNP Importance

1. In disease diagnosis
2. In finding predisposition to diseases
3. In drug discovery and development

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4. In drug responses

5. Investigation of migration patterns

6. Medication and diagnosis at individual level.

64. Write rule of mutation?

Mutation plays an important role in evolution. Mutation is important as the first step of evolution because it creates a new DNA sequence for a particular gene, creating a new allele. Recombination also can create a new DNA sequence (a new allele) for a specific gene through intergenic recombination.

65. Difference between chiasmata and crossing over?

chiasmata

Chiasmata formation during prophase I During pairing up of chromosomes, non sister chromatids exchange DNA segments. Each pair of chromosomes forms a tetrad, a group of four chromatids.

Crossing over

Crossing over is the swapping of genetic material that occurs in the germ line. During the formation of egg and sperm cells, also known as meiosis, paired chromosomes from each parent align so that similar DNA sequences from the paired chromosomes cross over one another.

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66.Meiosis I and meiosis II

Sister chromatids remained joined in meiosis I, whereas in mitosis they separate. In mitosis, centromere splits into two parts while in meiosis it does not splits.

No replication between Meiosis I and meiosis II: There is no chromosome replication between the end of meiosis I and beginning of meiosis II. Chromosomes are already replicated. Telophase I Haploid set of chromosomes arrived at each of the pole. Meiosis I result, a diploid cell produced two haploid cells. Prophase II of meiosis

Meiosis II similar to mitosis

- Meiosis II also occurs in four phases:
- Prophase II
- Metaphase II
- Anaphase II
- Telophase II and cytokinesis

67.Stages of Prophase I

Prophase I occupy more than 90% of the time required for meiosis. Five sub stages of prophase I

- Leptotene
- Zygotene
- Pachytene

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Diplotene

Diakinesis

68. Which cell division occur in somatic cells?

Mitotic cell division

69.Oogenesis and spermatogenesis

Meiosis in males is called spermatogenesis and produces sperms. Meiosis in females is called oogenesis and produces ova.

Spermatogenesis: Four sperm cells are produced from primary spermatocyte

Oogenesis: The polar bodies die. Only one ovum (egg) is produced from each primary oocyte.

70.Karyotype

Chromosomes are analyzed by organizing the into a karyotype. Karyotype is the general morphology of the somatic chromosome. Generally, karyotypes represent by arranging in the descending order of size keeping their centromere in a straight line.

71.Idiotype

Idiotype:

the karyotype of a species may be represented diagrammatically, showing all the morphological features of the chromosome; such a diagram is known as Idiotype

72.Difference between centromere and telomere?

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Centromere and telomeres are two essential features of all eukaryotic chromosomes. Each provide a unique function i.e., absolutely necessary for the stability of the chromosomes.

Centromeres are required for the segregation of the chromosomes during meiosis and mitosis.

Telomeres provide terminal stability to the chromosome and ensure its survival.

The region where two sister chromatids of a chromosome joined or "held together" during mitotic metaphase is called **centromere.**

When chromosomes stained, they show a dark-stained region that is the centromere. The region where two sister chromatids of a chromosome joined or "held together" During mitosis.

telomeres:

The two ends of a chromosome are known as telomeres. These are required for the replication and stability of the chromosomes

73.explain mitotic apparatus?

Mitotic apparatus/ spindles include

➢ Centrosomes

➢ Spindle microtubules

➢ Asters Mitotic apparatus/ spindles

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the apparatus of microtubules controls chromosome movement during mitosis.

74.<u>why high gene density in prokaryotes?</u>

Genome characteristics: Prokaryotes have less genome size as compared to eukaryotes.

Prokaryotes have a smaller number of genes as compared to eukaryotes. Prokaryotes have higher gene density as compared to eukaryotes

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